CAVS OFF-ROAD PROVING GROUND · AUTONOMOUS VEHICLE TESTING

Supervisory Monitor

Dashboard sketches · v4

Use case	Watch an autonomy run and stop the vehicle automatically when a listed trigger fires. The safety driver then takes over. Runs can be reviewed afterwards.
Where it runs	Web interface hosted on a CAVS server, opened in a desktop browser
Sensors in scope	The 3 vehicle cameras, plus LiDAR shown as a 2D occupancy grid or a 2D panorama image. No raw point cloud and no 3D view in the browser.
Backend	The provided ROS 2 bag autonomy_demo_0, played back as a mock live session. The interface shows the recording's own date and time (Fri 11 Sep 2026, 11:45:48–11:47:00 CDT), as if the run were happening now.
Pages	1 Live, normal · 2 Live, trigger fired · 3 Run review · 4 LiDAR view · 5 Trigger list · 6 Data budget · 7 Navigation

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1 · Dashboard, live — normal driving

Mock live session from the bag, Fri 11 Sep 2026 at 11:46:00 CDT. Autonomy active, every trigger armed.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor | autonomy_demo_0 | ● LIVE | Fri 11 Sep 2026 · 11:46:00 CDT | source: bag playback | AUTONOMY

STOP TRIGGERS — Edit list ›

- T1 Camera frame gap — > 158 ms (3 frames) — STOP armed
- T2 Camera rate drop — < 15 Hz — STOP armed
- T3 Camera feed lost — > 0.5 s — STOP armed
- T4 Localization stale — > 0.2 s — STOP armed
- T5 Autonomy command stale — autonomy_cmd_fresh = false — STOP armed
- T6 Path follower aborted — patience exceeded — ALERT armed
- T7 Collision monitor zone — slowdown or stop zone — ALERT armed

CAMERAS · PREVIEW 1 FRAME/S — Cameras ›

Cam 0 · 19 Hz | Cam 1 · 19 Hz | Cam 2 · 19 Hz

ROUTE · 2D — Open in Path Planning ›

1 Start

Goal (14.5, −12.3)

5 m

>1 m/s | 0.3–1 m/s | creeping

LIDAR · OCCUPANCY GRID — Grid | Panorama

obstacle
near obstacle
free

5 m

30 × 30 m around the vehicle
0.2 m cells · 1.7 updates/s
≈ 1.3 kB per update

EVENT LOG — Run Logs ›

11:45:56 Autonomy engaged, 2 waypoints

Mode: MANUAL

Speed

Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

A The clock shows the date and time the bag was recorded, advancing with playback, so the session reads as live.

B Trigger list on the left shows each stop condition and its live state. Green means armed and not met.

C Camera previews refresh once per second. LiDAR is shown as a 2D occupancy grid centred on the vehicle, never as a point cloud.

D Route is drawn in 2D from the vehicle pose only.

E No drive controls. A person in the vehicle can always take over.

2 · Dashboard, live — trigger fired

Same session at 11:46:06 CDT. Camera 2 goes 265 ms without a frame, so T1 fires.

cavs-server / hci-cavs1

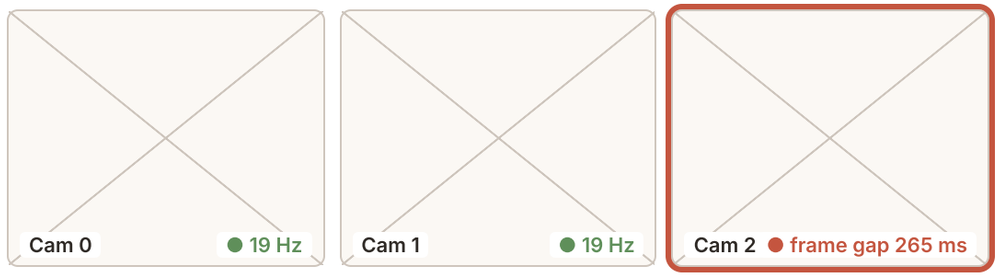
CAVS · Supervisory Monitor autonomy_demo_0 ▾ ● LIVE Fri 11 Sep 2026 · 11:46:06 CDT source: bag playback **STOPPED**

STOP sent · 11:46:06.0 T1 Camera frame gap: Camera 2, 265 ms (limit 158 ms). Vehicle halted, safety driver in control.

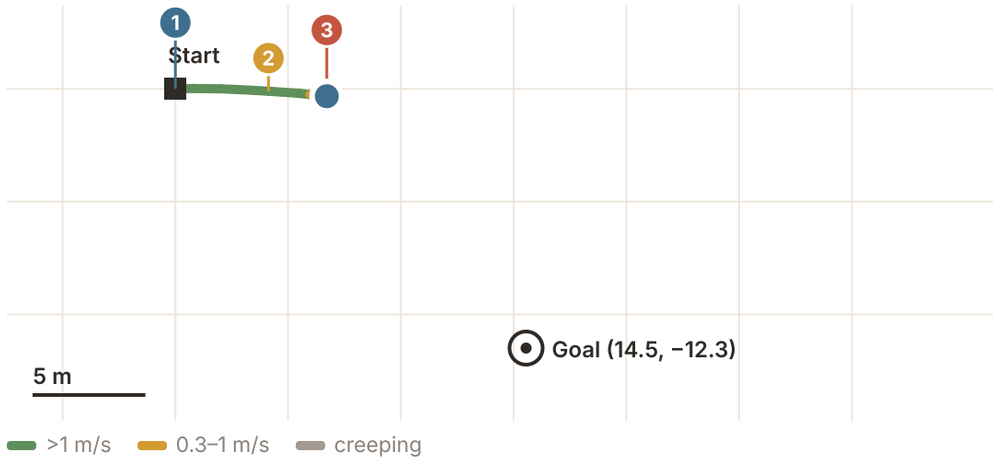
STOP TRIGGERS — Edit list ›

- **T1 Camera frame gap** — > 158 ms (3 frames) — STOP 265 ms
- **T2 Camera rate drop** — < 15 Hz — STOP armed
- **T3 Camera feed lost** — > 0.5 s — STOP armed
- **T4 Localization stale** — > 0.2 s — STOP armed
- **T5 Autonomy command stale** — autonomy_cmd_fresh = false — STOP armed
- **T6 Path follower aborted** — patience exceeded — ALERT at 11:46:03
- **T7 Collision monitor zone** — slowdown or stop zone — ALERT armed

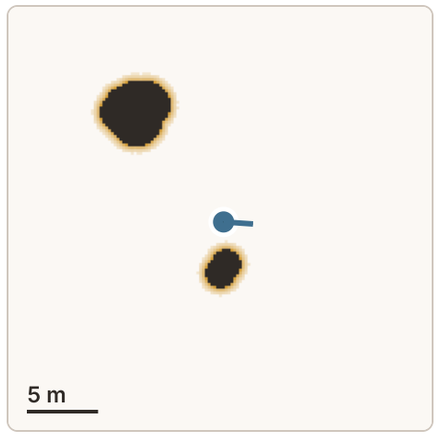
CAMERAS · PREVIEW 1 FRAME/S — Cameras ›



ROUTE · 2D — Open in Path Planning ›



LIDAR · OCCUPANCY GRID — Grid | Panorama



obstacle
near obstacle
free

30 × 30 m around the vehicle
0.2 m cells · 1.7 updates/s
≈ 0.7 kB per update

EVENT LOG — Run Logs ›

- 11:46:06 STOP sent on /mtx/estop · safety driver in control
- 11:46:06 T1 Camera 2 frame gap 265 ms
- 11:46:03 T6 Path follower aborted
- 11:45:56 Autonomy engaged, 2 waypoints



Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

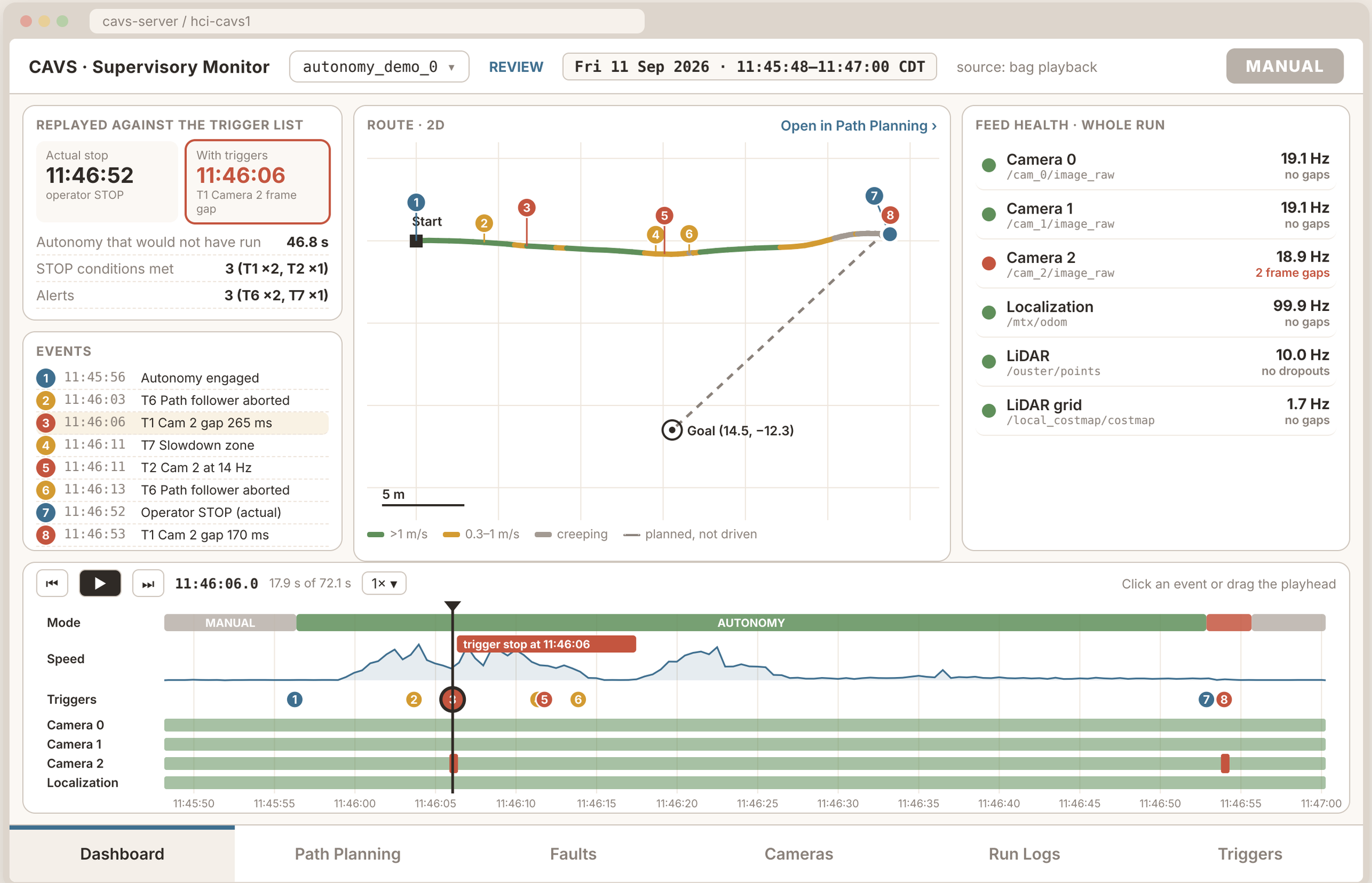
A A full-width banner states which trigger fired, the measured value against its limit, and that the stop was sent.

B The tripped trigger and camera are outlined in red. The other triggers keep their state.

C The stop is sent once and is not repeated. Resuming autonomy is done from the vehicle, not from this screen.

3 · Dashboard, review — after the run

The full run on one time axis, with the trigger list applied to the recorded data.



A Comparison card: the operator stopped this run at 11:46:52. With the trigger list, the stop would have come at 11:46:06, 46.8 s earlier.

B Timeline lanes are the mode, speed, trigger events and one lane per camera plus localization.

C Clicking an event moves the map, camera frames and log to that moment.

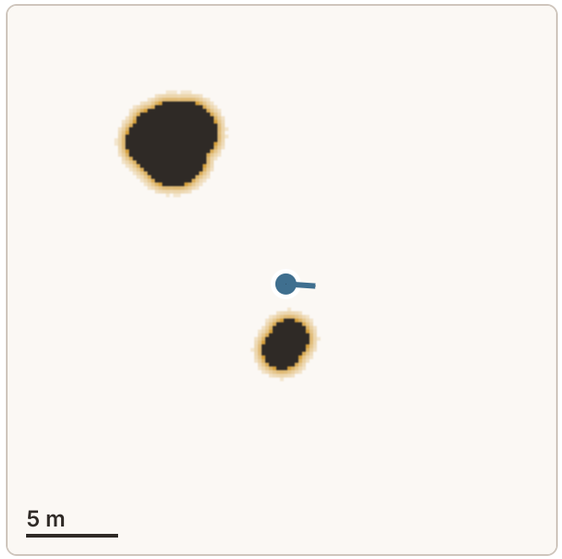
4 · LiDAR view

LiDAR reaches the browser only as a 2D image: an occupancy grid by default, or a panorama when the panel is switched.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor autonomy_demo_0 ▾ ● LIVE Fri 11 Sep 2026 · 11:46:06 CDT source: bag playback STOPPED

OCCUPANCY GRID **Grid** Panorama

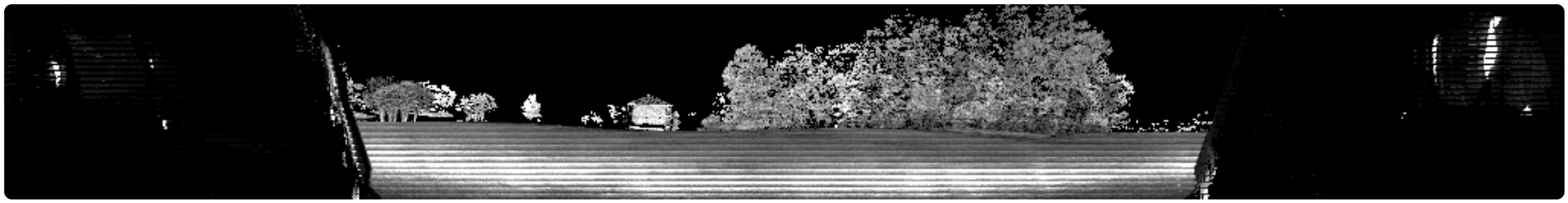


11:46:06 CDT · 30 × 30 m · 0.2 m cells

HOW EACH LIDAR VIEW IS MADE

VIEW	SOURCE TOPIC	ORIGIN	SERVER STEP	TO BROWSER
● **Occupancy grid**	`/local_costmap/costmap`	already built by Nav2 from the LiDAR	int8 grid → PNG, 0.2 ms	**≈ 0.7 kB × 1.7/s**
● **Panorama image**	`/ouster/reflec_image`	already built by the Ouster driver	16-bit → 8-bit JPEG, 0.5 ms	**≈ 22 kB × 1/s, only while open**
● **Raw point cloud**	`/ouster/points`		not sent	**6.3 MB × 10/s**

Both views reuse images the vehicle software already produces. The server only compresses them, so no point-cloud processing is added.



LiDAR panorama · reflectivity · 11:46:06 CDT · 1024 × 128

Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

A The grid is a top-down 30 × 30 m view centred on the vehicle. Dark cells are obstacles, amber cells are close to one.

B The panorama shows the full 360° LiDAR sweep as a flat image. It is sent only while someone has it open.

C Neither view needs the raw point cloud in the browser.

5 · Trigger list

The conditions that stop the vehicle. Thresholds are first proposals and can be tuned with CAVS.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor autonomy_demo_0 ▾ REVIEW Fri 11 Sep 2026 · 11:45:48–11:47:00 CDT source: bag playback MANUAL

STOP TRIGGER LIST

ID	TRIGGER	SIGNAL WATCHED	THRESHOLD	ACTION	FIRED IN AUTONOMY_DEMO_0
T1	Camera frame gap	`time between frames, any camera`	> 158 ms (3 frames)	STOP	first at 11:46:06 (17.9 s
T2	Camera rate drop	`frames per second, 1 s window`	< 15 Hz	STOP	first at 11:46:11 (23.6 s
T3	Camera feed lost	`no frame received`	> 0.5 s	STOP	
T4	Localization stale	`time since last /mtx/odom`	> 0.2 s	STOP	
T5	Autonomy command stale	`/mtx/arbiter_status`	autonomy_cmd_fresh = false	STOP	
T6	Path follower aborted	`controller_server log`	patience exceeded	ALERT	first at 11:46:03 (15.5 s
T7	Collision monitor zone	`/mtx/nav/collision_monitor_state`	slowdown or stop zone	ALERT	first at 11:46:11 (23.2 s

STOP publishes `/mtx/estop = true` once, from a trigger monitor node running beside the vehicle stack, so a stop does not depend on the network link. The vehicle halts and the safety driver takes over. The interface has no Go or resume control.

ALERT flags the event on the dashboard and in the log. Driving continues.

Every trigger is computed from message timing and small status topics. No image or point-cloud payload is needed to evaluate them.

Dashboard | Path Planning | Faults | Cameras | Run Logs | **Triggers**

A T1–T3 cover the three cameras. T4–T5 cover localization and the autonomy command link.

B T6–T7 are alerts only, since the navigation stack already slows or replans for them.

C The last column comes from replaying the provided bag.

6 · Data budget

Why the interface stays light: triggers are evaluated on the vehicle side, and only small summaries reach the browser.

cavs-server / hci-cavs1

CAVS · Supervisory Monitor | autonomy_demo_0 ▾ | REVIEW | Fri 11 Sep 2026 · 11:45:48–11:47:00 CDT | source: bag playback | MANUAL

ROS 2 bag playback
stands in for the vehicle now
+ trigger monitor node

→ **rosbridge**
filtered, throttled topics →

CAVS server
hosts the interface
keeps run history

→ **websocket**
≈ 50–75 kB/s →

Browser
any desktop
no install

STAYS ON THE VEHICLE SIDE

3 cameras, raw	2048 × 1536 mono8 · 19 Hz each	**≈ 179 MB/s**
Ouster LiDAR point cloud, raw	6.3 MB per scan · 10 Hz	**≈ 63 MB/s**
3D map / point cloud view		**not sent**

SENT TO THE BROWSER

Trigger states	from the trigger monitor node	**< 1 kB/s**
Vehicle pose for 2D map	/mtx/odom throttled to 5 Hz	**≈ 4 kB/s**
Status + log lines	/mtx/arbiter_status, /rosout	**≈ 2 kB/s**
Camera previews	3 × 320 × 240 JPEG · 1 frame/s	**≈ 45 kB/s**
LiDAR occupancy grid	150 × 150 PNG · 1.7 updates/s	**≈ 2 kB/s**
LiDAR panorama (when opened)	1024 × 128 JPEG · 1 frame/s	**≈ 22 kB/s**

≈ 50–75 kB/s to the browser against **≈ 240 MB/s** of raw camera and LiDAR data on the vehicle.

Dashboard | Path Planning | Faults | Cameras | Run Logs | Triggers

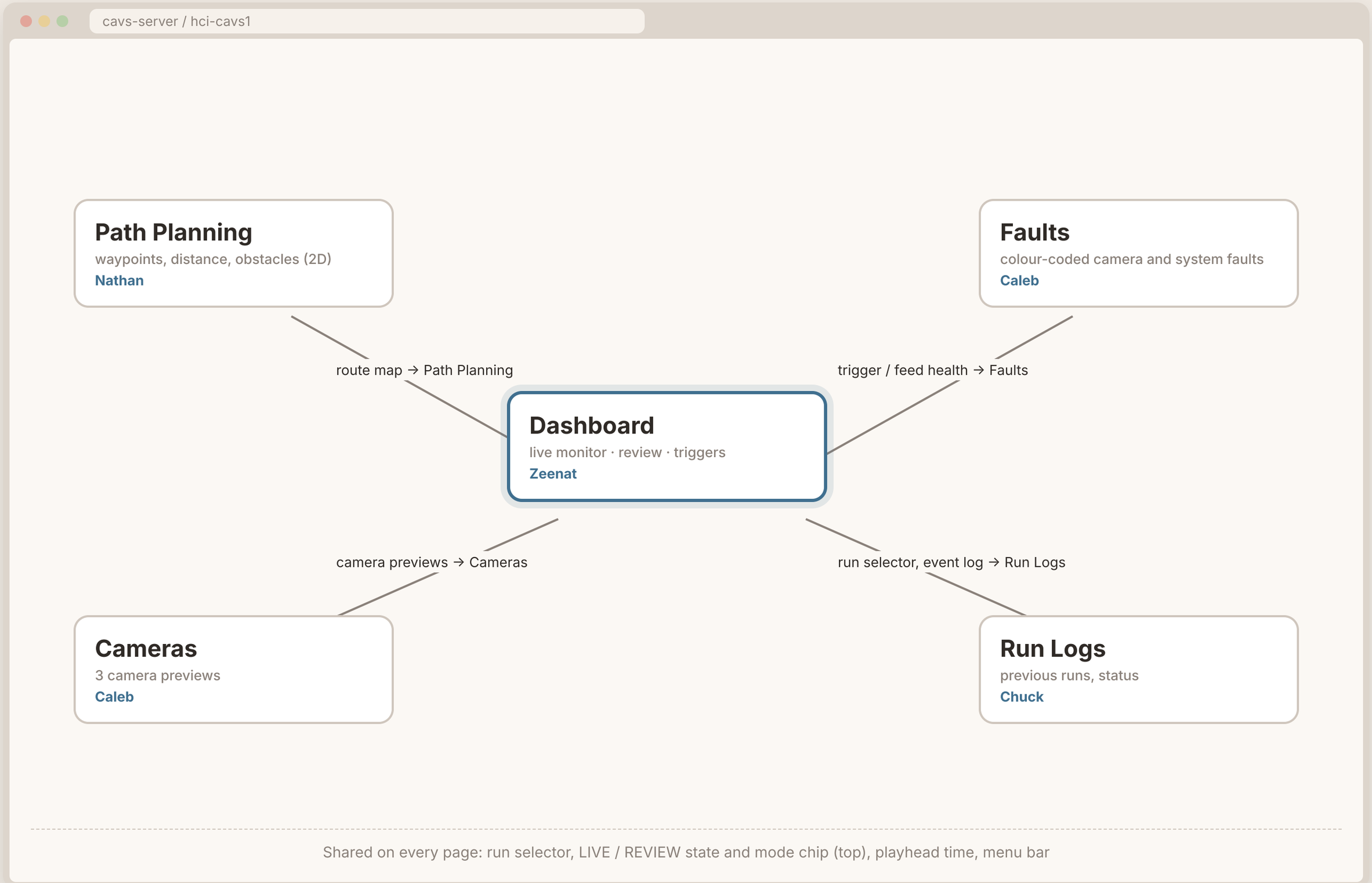
(A) Raw sizes are measured from the bag: each camera frame is 3.1 MB and each point cloud 6.3 MB.

(B) Previews and pose are throttled before they leave the server, so several people can watch at once.

(C) The same layout works with the recorded bag now and with the live vehicle later.

7 · Navigation

Where the dashboard hands off to the other pages in the team's design.



A The dashboard is the hub. Each panel links to the page that owns its detail.

B The selected run and playhead time carry over between pages.